

PREDICTIVE ALGORITHM STRATEGY

Part 1 & 2: Proof of Concept to Production Scale (ML Engine Selection)

Project: MNC HR Attrition Prediction Model

Objective: Select machine learning algorithms capable of answering point-in-time risk (Q1), time-horizon risk (Q2), and what-if scenario (Q3) queries while integrating seamlessly into our Databricks Dual-Layer Architecture.

1. THE BUSINESS QUERIES (WHAT HR IS ASKING)

These are the three distinct mathematical outputs the selected algorithms must generate from the 32 encoded attributes.

Query	Algorithm Action Required	Business Justification
Q1 (Point-in-Time)	Calculate a single, cumulative hazard score and convert to a 0-100% probability.	Provides a "General Flight Risk Score" used to rank the entire 50,000+ workforce from safest to riskiest on a dashboard.
Q2 (Time-Horizon)	Shift a baseline survival curve based on employee features, and output probabilities at exactly t=1, 2, 3, 6, 12 months.	Enables proactive retention planning. HR needs to know <i>when</i> to intervene (e.g., high risk at 6 months vs. high risk at 1 month).
Q3 (What-If Scenarios)	Alter a specific feature (e.g., MonthlyIncome) in the model and recalculate the Q2 time-horizon probabilities.	Calculates the ROI of interventions. Answers: "If we give this employee a 10% raise today, how much does their 6-month risk drop?"

2. ALGORITHM REJECTIONS (WHAT WE DO NOT USE)

While many algorithms can predict attrition, they fail to meet the requirements of Q1, Q2, and Q3 simultaneously.

Algorithm	Action / Capability	Business Justification for Rejection
Kaplan-Meier	Visual / Descriptive only.	Cannot ingest 32 parameters. Can only show survival curves for broad groups (e.g., "Sales vs. HR"), not individual employees.
Parametric Models (Weibull, Exponential)	Forces data into a strict, smooth mathematical curve.	HR behavior is messy; attrition happens in sudden "cliffs" (e.g., right after stock options vest). These models mathematically smooth over those cliffs, causing inaccurate Q2 predictions.
DeepSurv (Neural Networks)	Deep learning applied to survival analysis.	Acts as a "black box." When HR asks <i>why</i> an employee is flagged, we cannot explain the logic. Also, massive overkill for 32 tabular columns.
Aalen's Additive Model	Additive statistical modeling.	Mathematically unstable for generating clean, reliable 1, 3, 6, and 12-month business reports.
Basic Random Forest	Standard classification (Yes/No).	Can answer Q1, but completely incapable of answering Q2 (Time-Horizon probabilities) or Q3 (What-Ifs).

3. ALGORITHM SELECTIONS (THE DUAL-MODEL APPROACH)

To answer all business questions with both high accuracy and business trust, we deploy two specific algorithms in tandem within our Layer 2 Databricks compute.

Algorithm	Queries Answered	Business Justification
XGBoost Survival Models (Primary Engine)	Q1, Q2, Q3	The absolute most accurate algorithm for tabular HR data. It learns complex combinations (e.g., <i>High Overtime + Low Satisfaction</i>). It naturally supports "What-If" simulations by altering feature inputs and recalculating time-horizons. Natively optimized in Databricks.
Cox Proportional Hazards (Secondary Engine)	Q1, Q2, Q3 (Specifically the "Why")	Acts as our "glass box" validator. While XGBoost is accurate, Cox PH provides exact mathematical coefficients. It is the <i>only</i> algorithm that can definitively answer: " <i>Of our 5 Logical Themes, Sentiment accounts for 42% of the risk, while Compensation accounts for 11%.</i> " Used to justify XGBoost predictions to HR.
Multi-State Models (Future Phase 3)	Complex Paths (Promotion -> Attrition)	Not used in Part 1 or 2. Reserved for future roadmap where we predict multi-step employee journeys rather than just binary attrition.

4. ARCHITECTURE INTEGRATION & OUTPUT MAPPING

Integrating the XGBoost Survival model requires no structural changes to our Layer 1 Delta Tables. We only expand the write-back columns pushed from Layer 2 to Layer 1.

Output Column / Action	Algorithm Source	Business Justification
Prob_Leave_1M	XGBoost Survival (t=1)	Flags employees in immediate, acute danger of leaving. Used for urgent HR interventions.
Prob_Leave_3M	XGBoost Survival (t=3)	Identifies short-term flight risks.
Prob_Leave_6M	XGBoost Survival (t=6)	The primary "What-If" column. HR simulates raises/ interventions and checks if this 6-month score drops below the danger threshold.
Prob_Leave_12M	XGBoost Survival (t=12)	Long-term strategic workforce planning.
General_Risk_Score	Cox PH / XGBoost	A single 0-100 ranking used by executives to filter and sort the entire company dashboard.
Theme Risk Percentages	Cox PH Coefficients	A separate dashboard metric explaining which of the 5 Themes (Compensation, Sentiment, etc.) is driving the General_Risk_Score.

5. EXCLUSIONS & MODEL DATA HANDLING

- **Excluded Target Variable Handling:** The Attrition column (Yes=1, No=0) is not treated as a standard classification label. For Survival algorithms, it is paired with the YearsAtCompany column to create a "Time-to-Event" structure, telling the math *when* the event occurred, not just *if* it occurred.
- **Excluded Feature Set:** The 3 dropped constant columns (EmployeeCount, StandardHours, Over18) are never fed to these algorithms, ensuring compute resources are strictly focused on predictive signals.